

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and optimized **large-scale AI model pre-training pipelines** on distributed GPU infrastructures, accelerating compute efficiency by **20%**.
- Developed and implemented novel **transformer architectures** for large language models, enhancing predictive accuracy by **15%**.
- Executed research experiments for **model scaling and performance tuning**, identifying and resolving computational bottlenecks that reduced training time by **18%**.
- Enhanced **distributed training systems** for reliability and scalability, ensuring seamless operation across compute environments handling over **10TB** of training data.
- Contributed to data curation strategies and baseline improvements for model training, strengthening outcomes and reducing data processing latency by **25%**.
- Collaborated with cross-functional engineering and research teams to accelerate innovation, integrating new methodologies into existing **AI training systems**.

SKILLS

AI/ML Frameworks – PyTorch, TensorFlow, JAX

Model Architectures – Transformer, Large Language Models (LLMs), Multi-modal AI

Distributed Systems – Distributed Training, Parallel Training Strategies, GPU Compute

Programming Languages – Python, SQL

Data Engineering & Analytics – Data Curation, ETL, Data Warehousing

Cloud & Infrastructure – AWS, Docker, Linux

ML Operations – Model Optimization, Performance Tuning, Scalability

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

**BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India**

01/2021

PROJECTS

Distributed Training Pipeline for Natural Language Understanding Models

- Engineered a distributed training pipeline using **PyTorch** and **AWS** services (EC2, S3) to train large-scale language models.
- Implemented parallel training strategies and optimized GPU utilization, achieving a **30%** reduction in training cycles for a 100B parameter model.
- Developed custom data loaders and pre-processing scripts in **Python** to handle terabytes of text data efficiently.

AI Research Platform for Multi-Modal Model Experimentation

- Built a research platform for experimenting with novel **transformer architectures** for multi-modal AI systems.
- Integrated **TensorFlow** and **JAX** for flexible model development and research, supporting rapid iteration and hypothesis testing.
- Designed experiments to analyze model performance and scalability on large GPU clusters, contributing to baseline improvements for future research initiatives.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)